

1) Design a simple HTML webpage that displays:

- **A main heading**
- **A paragraph with customized font and color**
- **A clickable link**

Write the appropriate HTML code.

ANS :-

```
<!DOCTYPE html>
<html>

<head>
<title>My Webpage</title>
</head>

<body>

<!-- A main heading -->
<h1 style="color: blue; font-family: Arial;">Welcome</h1>

<p style="color: green; font-family: Verdana;">
This is a sample webpage.
</p>

<br>

<a href="https://example.com">Visit Website</a>

</body>
</html>
```

**2) Create an HTML form to collect user details such as name, email, and gender.
Use appropriate form elements.**

ANS :-

```
<!DOCTYPE html>
<html>
<head>
<title>Form</title>
</head>
<body>

<form>

Name: <input type="text" name="name"><br><br>
```

Email: <input type="email" name="email">

Gender:

<input type="radio" name="gender"> Male

<input type="radio" name="gender"> Female

<input type="submit" value="Submit">

</form>

</body>

</html>

3) You are required to display a list of items on a webpage in bullet form. Which HTML element will you use? Write a simple example

ANS :-

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Bullet List Example</title>
```

```
</head>
```

```
<body>
```

```
<h2>Fruits List</h2>
```

```
<ul>
```

```
<li>Apple</li>
```

```
<li>Banana</li>
```

```
<li>Mango</li>
```

```
<li>Orange</li>
```

```
</ul>
```

```
</body>
```

```
</html>
```

4) Using JavaScript, change the background color of a paragraph to yellow when a button is clicked. Write the HTML and JavaScript code.

ANS :-

```
<!DOCTYPE html>
```

```

<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>JS Background Changer</title>
  <style>
    body { font-family: sans-serif; padding: 50px; text-align: center; }
    #target-paragraph {
      padding: 20px;
      border: 1px solid #ddd;
      transition: background-color 0.3s ease; /* Smooth transition */
    }
    button { padding: 10px 20px; cursor: pointer; background: #34495e; color: white; border: none;
border-radius: 5px; }
  </style>
</head>
<body>

  <h2>JavaScript Event Handling</h2>

  <p id="target-paragraph">
    Click the button below to highlight this text with a yellow background!
  </p>

  <button onclick="changeColor()">Highlight Paragraph</button>

  <script>
    // This function runs when the button is clicked
    function changeColor() {
      // 1. Find the element by its unique ID
      const paragraph = document.getElementById('target-paragraph');

      // 2. Change the background color style property
      paragraph.style.backgroundColor = "yellow";
    }
  </script>

</body>
</html>

```

5) Create a JavaScript array of 5 numbers. Write a function that doubles each number and logs the result using a loop.

ANS :-

```
// 1. Create an array of 5 numbers
const myNumbers = [10, 25, 5, 42, 8];

// 2. Define the function to double and log numbers
function doubleAndLog(numbersArray) {
  console.log("Starting the doubling process...");

  // 3. Use a 'for' loop to iterate through the array
  for (let i = 0; i < numbersArray.length; i++) {
    let doubledValue = numbersArray[i] * 2;
    console.log("Original: " + numbersArray[i] + " -> Doubled: " + doubledValue);
  }

  console.log("Process complete!");
}

// 4. Call the function
doubleAndLog(myNumbers);
```

6) Design a responsive webpage for a student registration form using:

- Inline, Embedded, and External CSS (demonstrate rule cascading clearly)
- Apply backgrounds, borders, colors, and shadows
- Use text transformations (uppercase, capitalize, etc.)
- Add a hover transition effect on the submit button

Apply appropriate CSS3 properties and explain how cascading and inheritance are working in your design.

ANS :-

1. External CSS (style.css)

Save this as a separate file. This layer sets the **base defaults** that are inherited by the rest of the page.

CSS

```
/* External CSS:
Setting the foundation */
```

```
body {

  font-family: 'Segoe UI', sans-serif;
```

```

        background: #f0f2f5;
        margin: 0;
        padding: 20px;
        color: #333; /* This will be inherited by
most child elements */
    }
    .container {
        max-width: 500px;
        margin: auto;
    }

```

2. The HTML & Embedded CSS (index.html)

This file contains the structure, the page-specific styles (Embedded), and the direct overrides (Inline).

HTML

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Student Registration</title>
    <link rel="stylesheet" href="style.css">
    <style>
        /* Embedded CSS: Specific to this form
*/
        .form-card {
            background: #ffffff;
            padding: 30px;
            border-radius: 15px;
            border: 2px solid #3498db;
            box-shadow: 0 10px 25px rgba(0,0,0,0.1);
/* CSS3 Shadow */
        }
        h2 {
            text-align: center;
            text-transform: uppercase; /* Text
Transformation */

```

```

        color: #2c3e50;
        letter-spacing: 2px;
    }
    label {
        display: block;
        margin-top: 15px;
        text-transform: capitalize; /* Text
Transformation */
        font-weight: 600;
    }
    input {
        width: 100%;
        padding: 10px;
        margin-top: 5px;
        border: 1px solid #ddd;
        border-radius: 5px;
        box-sizing: border-box;
    }
    /* Submit Button with Transition */
    .btn-submit {
        margin-top: 25px;
        width: 100%;
        padding: 12px;
        background-color: #3498db;
        color: white;
        border: none;
        border-radius: 5px;
        cursor: pointer;
        font-size: 16px;
        text-transform: uppercase;
        transition: background-color 0.4s
ease, transform 0.2s; /* CSS3 Transition */
    }
    .btn-submit:hover {
        background-color: #2980b9;
        transform: translateY(-2px); /*
Hover effect */
        box-shadow: 0 5px 15px rgba(52, 152,
219, 0.4);
    }
</style>
</head>
<body>
    <div class="container">

```

```

    <div class="form-card">
    <h2 style="color: #e67e22;
border-bottom: 2px solid #e67e22;">Student Portal</h2>
    <form>
    <label for="fname">full
name</label>
    <input type="text"
id="fname" placeholder="Enter name...">
    <label for="email">email
address</label>
    <input type="email"
id="email" placeholder="student@university.edu">
    <label for="course">course
selection</label>
    <input type="text"
id="course" placeholder="e.g. Computer Science">
    <p style="font-size:
12px; color: #7f8c8d; font-style: italic;">
    * All fields are required
for registration.
    </p>
    <button type="submit"
class="btn-submit">Register Now</button>
    </form>
    </div>
    </div>
</body>
</html>

```

1. The Cascade (The "C" in CSS)

The browser resolves conflicts between rules based on **Specificity**. In this design:

-
- **External vs. Embedded:** If style.css said the background was red, but the <style> tag in the HTML said it was white, the **Embedded** style wins because it appears later in the document.
- **Inline Override:** Notice the <h2>. Even though the Embedded CSS says headings should be blue (#2c3e50), the **Inline** style (style="color: #e67e22") takes absolute priority because it is the most specific.

2. Inheritance

Inheritance is the process by which properties are passed from parent to child.

- In style.css, I set the font-family on the body.
- Notice I didn't define a font for the <p> tag or the <label> tags. They **inherited** "Segoe UI" automatically from the body. Properties like color, font, and line-height typically inherit, while border, margin, and padding do not.

3. CSS3 Enhancements

- **Transitions:** The transition property on the button creates a smooth "fade" between the blue and dark blue colors rather than an instant flick.
- **Box Shadow:** The box-shadow on .form-card creates depth, making the form look like it is floating above the background.
- **Responsiveness:** By using max-width: 500px and width: 100%, the form will shrink automatically when viewed on a mobile phone.

7) Create a horizontal navigation bar with dropdown menu for a college website using:

- **Advanced CSS concepts:** Grouping selectors, positioning, floating, alignment
- **Use pseudo-classes** (:hover, :active, :focus)
- **Implement image sprites for menu icons**
- **Add a smooth animation or transition effect when hovering over menu items**

Apply proper styling to make it visually attractive and responsive.

ANS :-

Creating a navigation

bar with a dropdown menu requires careful use of **CSS Positioning**. We use relative positioning on the parent item to create a "container" and absolute positioning on the dropdown so it can float over the content below it.

To handle the **Image**

Sprites, we use one single image file containing all icons and shift the background-position to show only the portion we need. This reduces server requests and speeds up the site.

The College Navigation Bar Code

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>College Navigation</title>
  <style>
    /* 1. Grouping Selectors & Reset */
    nav ul, nav li, nav a {
      margin: 0;
      padding: 0;
      list-style: none;
      text-decoration: none;
      box-sizing: border-box;
    }
    /* 2. Main Navigation Styling */
    nav {
      background-color: #002147; /*
Oxford Blue */
      height: 60px;
      font-family: 'Arial', sans-serif;
      border-bottom: 4px solid #f1c40f; /*
Gold accent */
    }
    /* 3. Floating and Alignment */
    nav > ul > li {
```

```

float: left; /* Using Floating as
requested */
position: relative; /* Essential
for dropdown positioning */
}
nav a {
display: block;
padding: 0 25px;
line-height: 60px;
color: white;
font-weight: bold;
transition: all 0.3s ease; /*
Smooth transition */
}
/* 4. Pseudo-classes: Hover, Active,
Focus */
nav a:hover, nav a:focus {
background-color: #003366;
color: #f1c40f;
}
nav li:active > a {
transform: scale(0.95);
}
/* 5. Image Sprite Implementation */
.icon {
display: inline-block;
width: 20px;
height: 20px;
vertical-align: middle;
margin-right: 8px;
/* Placeholder sprite map */
background: url('https://via.placeholder.com/100x20&#39;);
no-repeat;
}
/* Shifting background to show
different icons */
.icon-home { background-position: 0 0;
}
.icon-about { background-position: -25px
0; }
.icon-dept { background-position: -50px
0; }
/* 6. Dropdown Menu Styling */
.dropdown {

```

```

    position: absolute;
    top: 60px;
    left: 0;
    width: 200px;
    background: #ffffff;
    box-shadow: 0 8px 16px rgba(0,0,0,0.2);
    /* Hide by default for animation */
    opacity: 0;
    visibility: hidden;
    transform: translateY(10px);
    transition: opacity 0.4s, transform
0.4s;
    z-index: 100;
  }
  .dropdown li {
    width: 100%;
    border-bottom: 1px solid #eee;
  }
  .dropdown a {
    color: #333;
    line-height: 45px;
    padding: 0 15px;
  }
  /* 7. Show Dropdown on Hover */
  nav li:hover .dropdown {
    opacity: 1;
    visibility: visible;
    transform: translateY(0);
  }
  /* Responsive Fix */
  @media (max-width: 768px) {
    nav { height: auto; }
    nav > ul > li { float: none; width:
100%; }
    .dropdown { position: static; width:
100%; display: none; }
    nav li:hover .dropdown { display:
block; }
  }
</style>
</head>
<body>
<nav>
  <ul>

```

```

    </li>
    <a href="#"><span
class="icon icon-home"></span>Home</a>
  </li>
  <li>
    <a href="#"><span
class="icon icon-about"></span>About Us</a>
  </li>
  <li>
    <a href="#"><span class="icon
icon-dept"></span>Departments ▼</a>
    <ul class="dropdown">
      <li><a href="#">Computer
Science</a></li>
      <li><a href="#">Engineering</a></li>
      <li><a href="#">Arts
& Sciences</a></li>
      <li><a href="#">Business
School</a></li>
    </ul>
  </li>
  <li>
    <a href="#focus-test" id="focus-test">Admissions</a>
  </li>
</ul>
</nav>
<div style="padding:
50px; text-align: center;">
  <h1>Welcome to University Portal</h1>
  <p>Hover over "Departments"
to see the smooth dropdown effect.</p>
</div>
</body>
</html>

```

Key Concepts

Breakdown

Image Sprites

Instead of loading

four different PNG files for your menu icons, you load one "map." In the code, .icon-about uses background-position: -25px 0;, which slides the

background image to the left to reveal the second icon in the row. This is a classic performance-heavy technique.

Positioning & The "Anchor"

- **position: relative:** We apply this to the parent . It acts as the anchor point for the dropdown.
- **position: absolute:** We apply this to the .dropdown. This allows the menu to "break out" of the normal document flow and sit on top of other content without pushing other elements down.

Transitions and Visibility

Rather than using display: none, which isn't animatable, I used a combination of opacity: 0 and transform: translateY(10px). When you hover, the menu fades in and "slides" up into place, giving it a modern, premium feel.